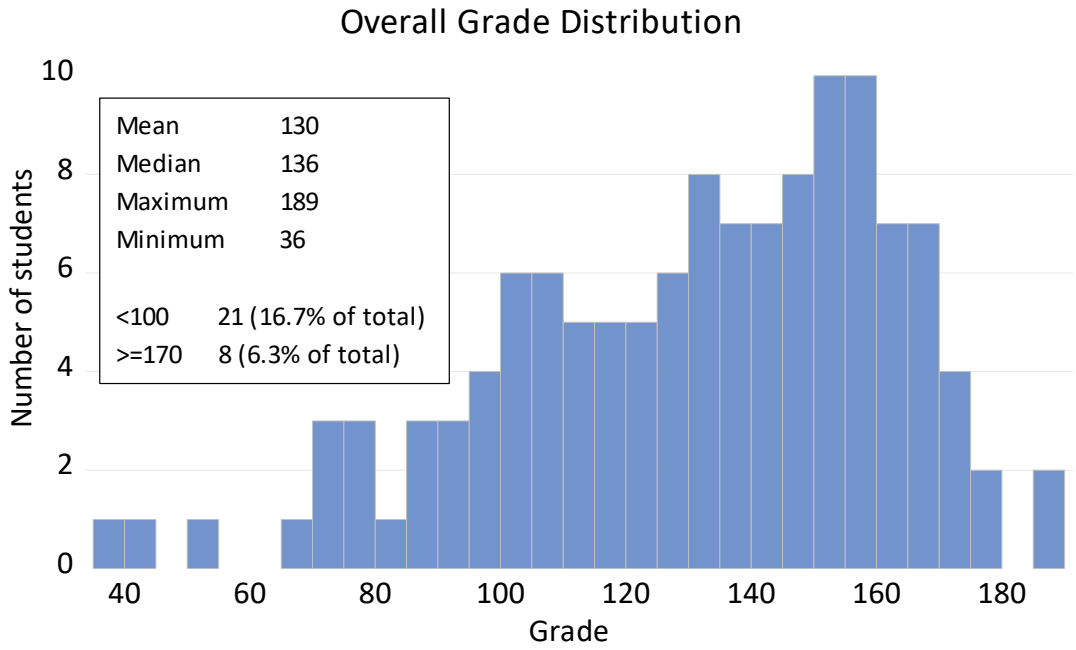
Midterm Feedback

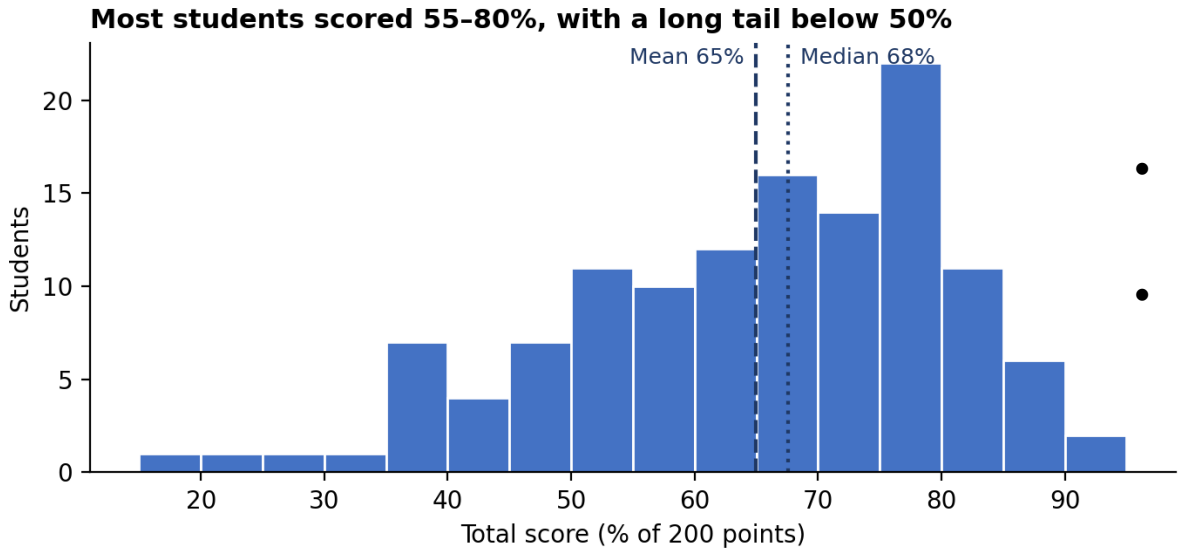
Econometrics 2026/2027

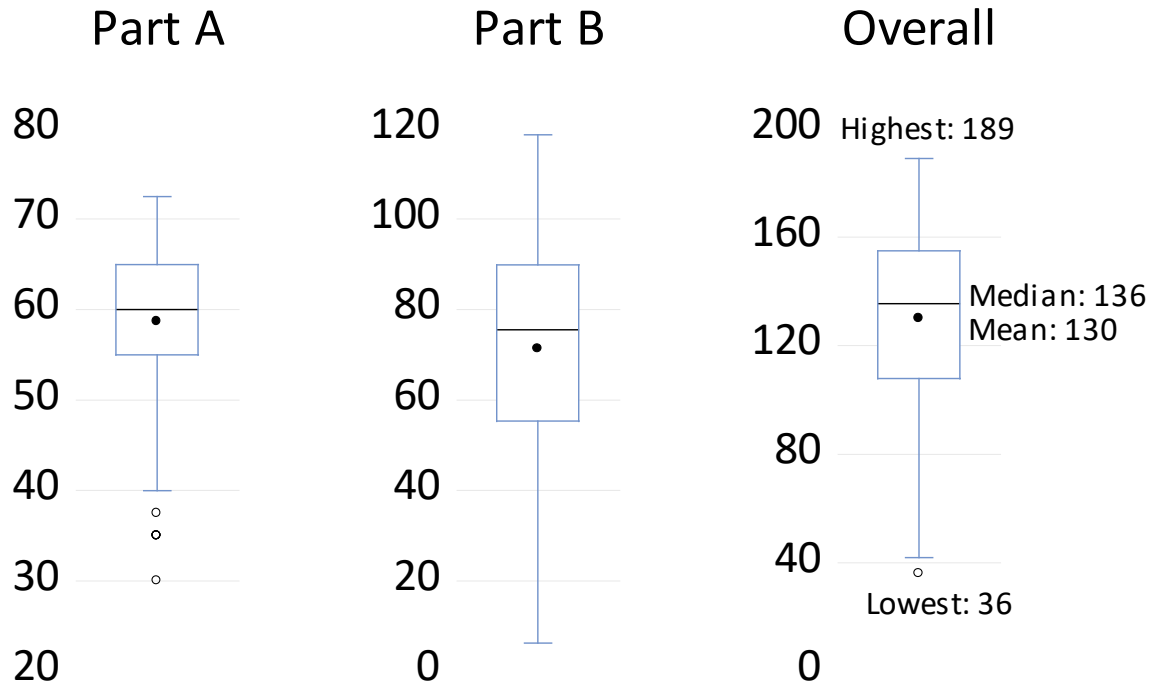
Key messages

- **The class averaged 65%.** Mean 130 of 200 points, median 136 points. 21 students (17%) scored less than 50%, and 19 (15%) scored above 80%.
- **The open questions decide the ranking.** Students averaged 81% on multiple choice but 56% on open questions, and 123 of 126 scored a lower percentage on Part B. Part B accounts for about three quarters of the variation in total scores.
- **Partial credit is the norm.** On Q30, Q39, Q40 and Q41, about 80% of students received partial credit and no more than 7% full credit. Students recognise the concept but rarely complete the argument.
- **The endogeneity/IV block (Q38–Q41) is the weakest area.** It carries 37% of Part B points but accounts for 52% of the Part B points lost.
- **Q38 is an outlier.** It averaged 19%, 59% of students scored zero, and even the top quarter averaged only 29%
- **Easy points were lost on Q36.** 61% of students gave at least one DiD answer as a bare number without the expression.

Overall results: Average grade is 130/200 (65%)

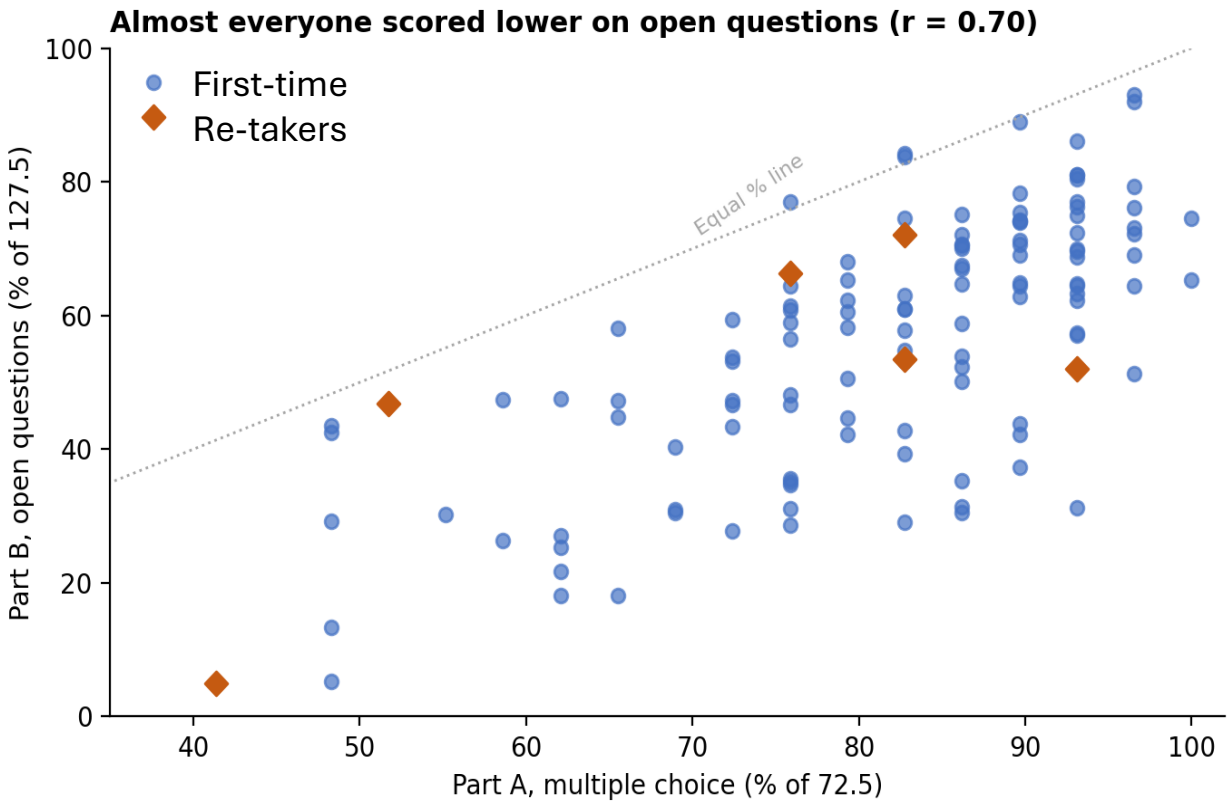


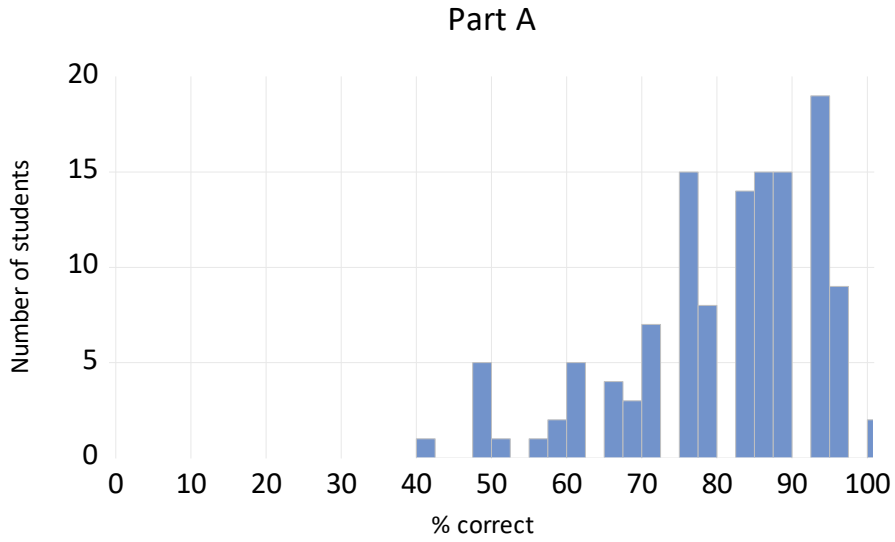


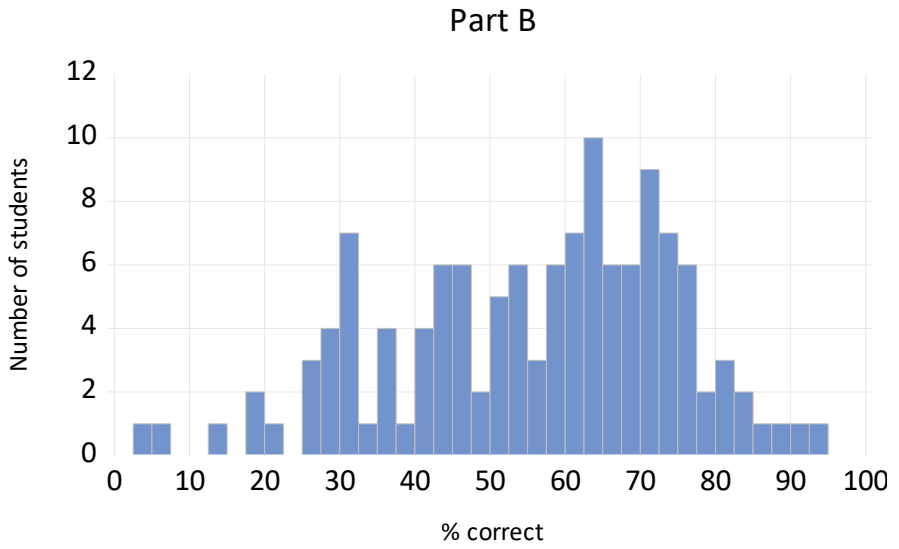


- **The class averaged 65%.** Mean 130 of 200 points, median 136.
- 21 students (17%) scored less than 100 points (<50%), and 19 (15%) scored above 80%.

Multiple choice vs. open questions

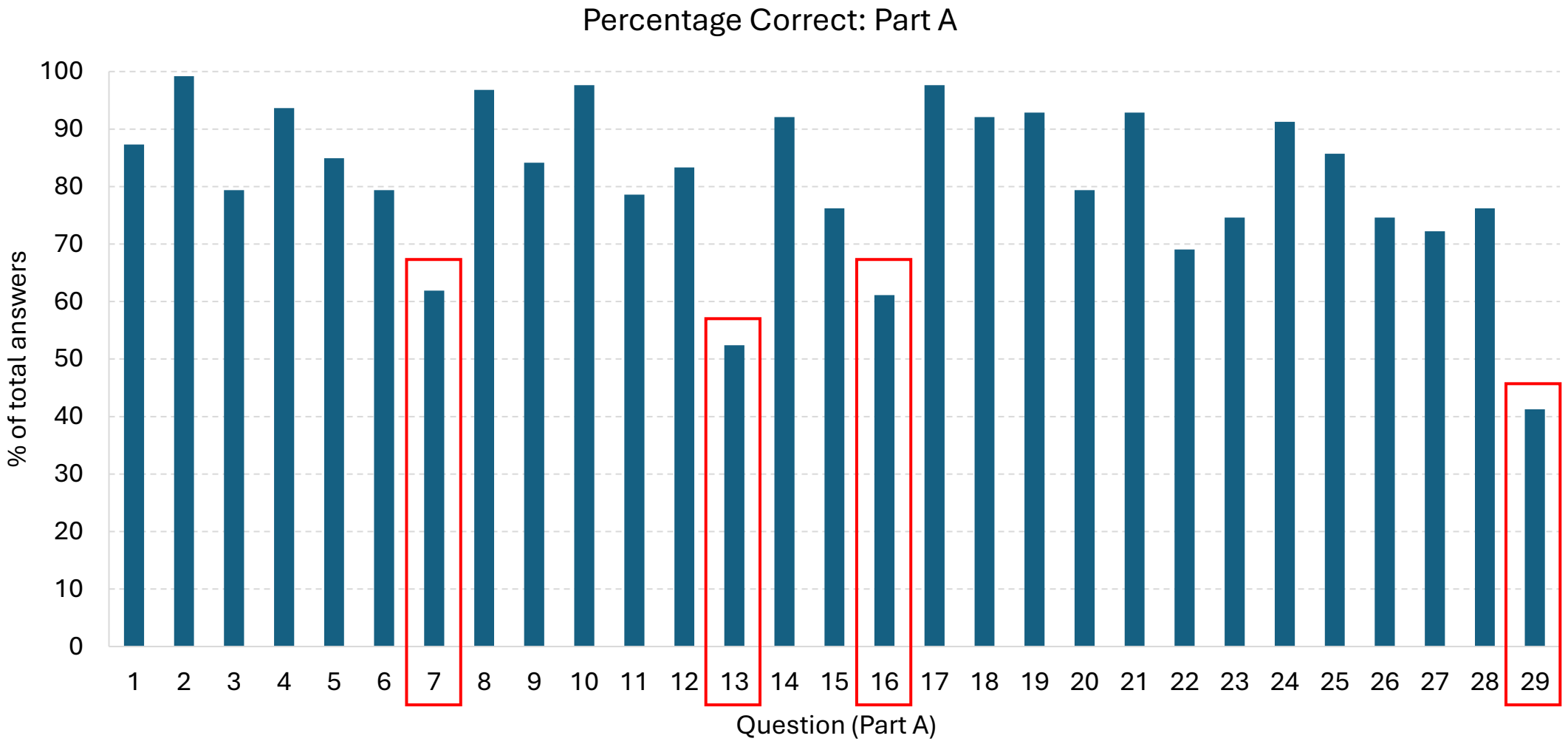






- **The open questions decide the ranking.** Students averaged 81% on multiple choice but 56% on open questions, and 123 of 126 scored a lower percentage on Part B. Part B accounts for about three quarters of the variation in total scores.

Part A: the average item is answered correctly by 81% of students



Part A: questions that caused difficulties (1)

7. Which of the following is a nonlinear regression model?

a. $y = \beta_0 + \beta_1 x^{1/2} + u$
b. $\log y = \beta_0 + \beta_1 \log x + u$
c. $y = 1 / (\beta_0 + \beta_1 x) + u$
d. $y = \beta_0 + \beta_1 x + u$

	% of answers
A	25 %
B	12 %
C	62 %
D	1 %

13. The significance level of a test is:

a. The probability of rejecting the null hypothesis when it is true
b. The probability of rejecting the null hypothesis when it is false
c. One minus the probability of rejecting the null hypothesis when it is false
d. One minus the probability of rejecting the null hypothesis when it is true

	% of answers
A	66 %
B	24 %
C	13 %
D	22 %

Part A: questions that caused difficulties (2)

16. In a regression model, if variance of the dependent variable, conditional on the explanatory variables, is not constant, then the:

a. *t*-statistics and confidence intervals are both invalid no matter how large the sample size is
b. *t*-statistics are invalid and confidence intervals are valid for small sample sizes
c. *t*-statistics are valid and confidence intervals are invalid for small sample sizes
d. *t*-statistics and confidence intervals are valid no matter how large the sample size is

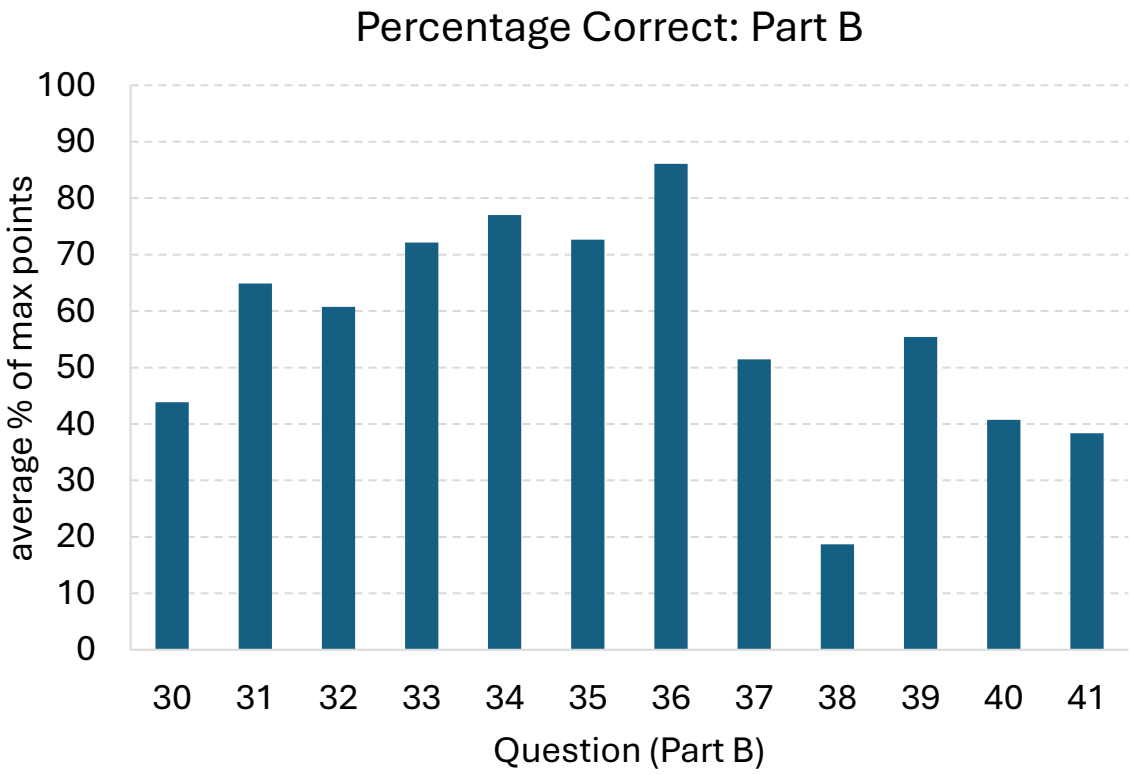
	% of answers
A	62 %
B	14 %
C	19 %
D	5 %

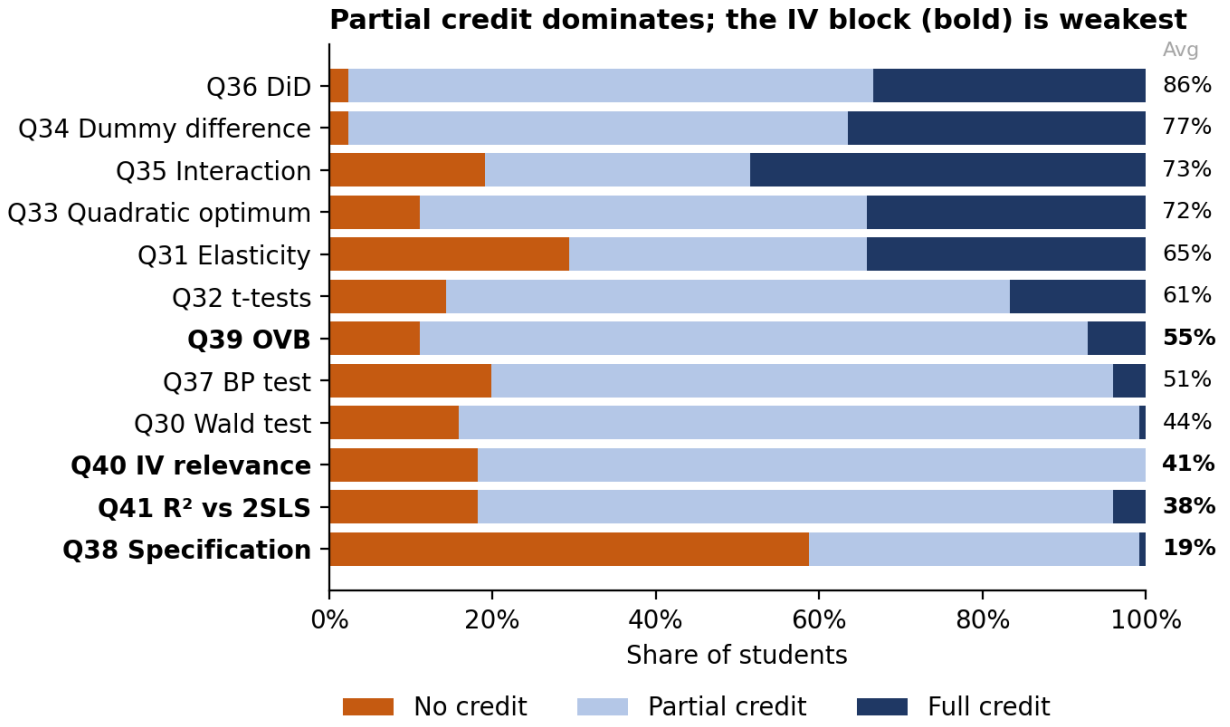
29. The order condition for identification of an equation (2SLS) requires that there should be _____

a. At least one exogenous explanatory variable in a structural equation
b. At least as many dummy variables in an equation as there are exogenous explanatory variables
c. As many lagged independent variables in an equation as there are exogenous explanatory variables
d. At least as many excluded exogenous explanatory variables as there are included endogenous explanatory variables

	% of answers
A	40 %
B	6 %
C	12 %
D	41 %

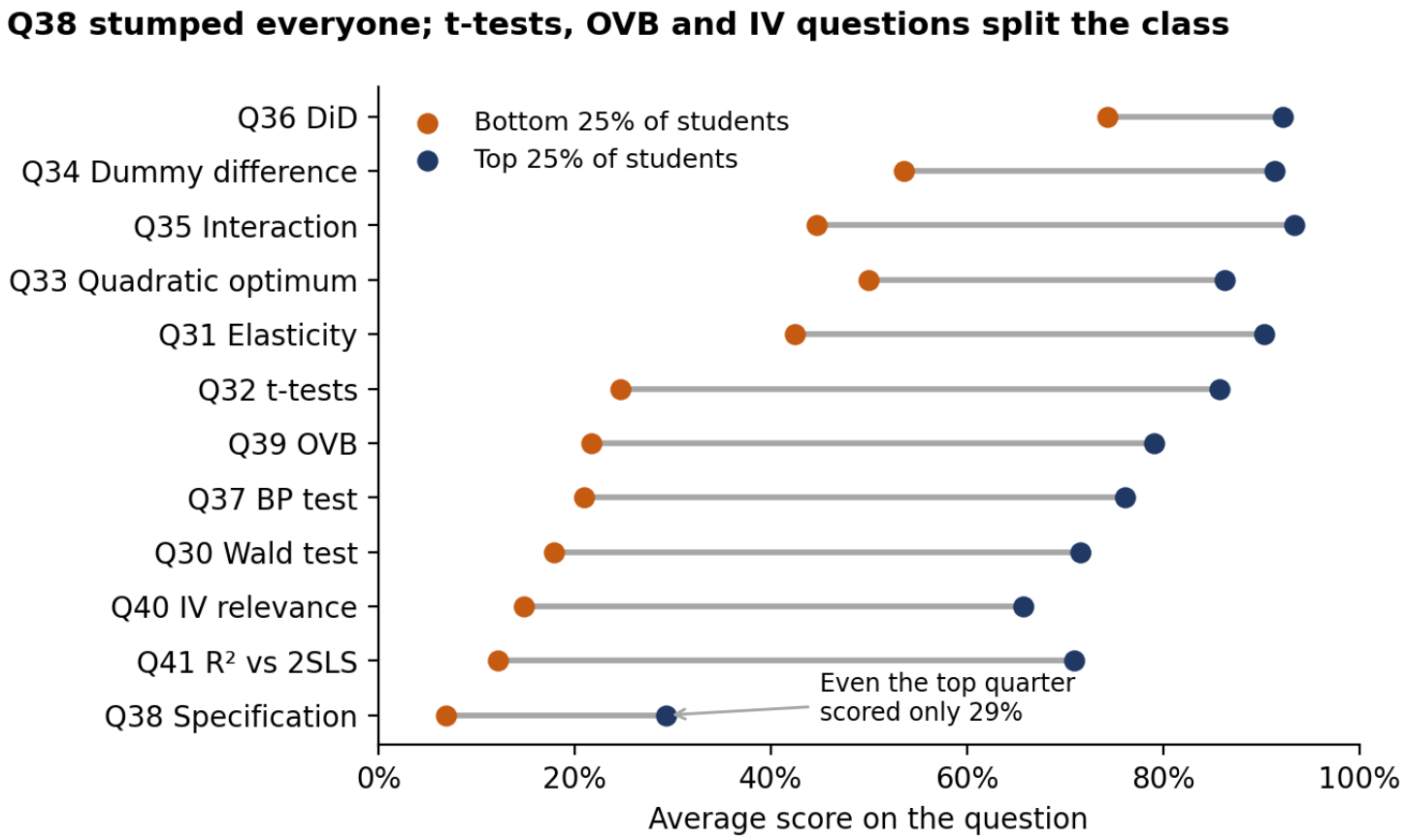
Part B: On average a student lost 56 of the 127.5 Part B points





- **The endogeneity/IV block (Q38–Q41) is the weakest area.** It carries 37% of Part B points but accounts for 52% of the Part B points lost.

What separates students



The gap between the top and bottom quarters is widest on Q32 (t-tests), Q41, Q39 and Q37, at 55–61 percentage points. These questions reward a complete chain of reasoning.

Q38 shows the opposite pattern: the gap is small because nearly everyone failed it.

Q30: Wald test and multicollinearity. Only 1 student earned full credit

A complete answer. $p = 0.0304 < 0.05$, so we reject H_0: $\beta_{LINCOME} = \beta_{PRPPOV} = 0$ at 5% level. Individually, both are insignificant ($p = 0.16$ and 0.70). Income and the poverty share are strongly correlated, which inflates their standard errors. The individual t-statistics fall, while the joint test still detects their combined effect.

Common mistakes.

- Reading the test as "not significant", or reporting only the 1% level and missing that the variables are jointly significant at 5%.
- Stating the contrast without diagnosing it, or naming multicollinearity without the contrast that supports it.
- Wrong diagnoses: low R^2, omitted variable bias, functional form misspecification, or the variables being *un*correlated.
- Wrong mechanisms: "the coefficients cancel each other out". The mechanism is inflated standard errors.
- Wrong remedies. An interaction term adds a regressor correlated with both and makes the problem worse.
- Misidentifying the variables.

Q31: Elasticity.
Results are polarised: 29% scored zero and 34% full marks

Complete answer. The correct version: a 10% increase in population is associated with about a 0.66% increase in rent, ceteris paribus. (also accepted: “A 1% increase … → … 0.066%, …”).

Common mistakes.

- "A 1% increase is associated with a 6.6% increase": the same factor-of-100 error.
- "A 1 unit increase in population": the elasticity is per 1%, not per person.
- Forgotten "other factors held constant”.

Q32: t-test.
The gap between the top and bottom quarters is the widest

Complete answer.

For pctstu, H_0: $\beta_{pctstu} = 0$, H_1: $\beta_{pctstu} \neq 0$.

t-stat = $\hat{\beta}_{pctstu}/se(\hat{\beta}_{pctstu}) = 0.0056/0.0017 \approx 3.29 > 2$, so reject H_0 at 5% significance level.

The size of the student body relative to the population is significant at 5% level.

For log(pop), H_0: $\beta_{log(pop)} = 0$, H_1: $\beta_{log(pop)} \neq 0$.

t-stat = $\hat{\beta}_{log(pop)}/se(\hat{\beta}_{log(pop)}) = 0.066/0.039 \approx 1.69 < 2$, so fail to reject H_0 at 5% significance level.

The total city population is not significant at 5% level.

Common mistakes.

- Decisions without the t-statistics.
- Using 1.645 as the critical value. That is one-sided, so it wrongly rejects for log(pop).
- Reversing the decision rule, or a decimal slip (t = 16.9) that flips the log(pop) decision.
- "Accept H_0" or "population has no effect" instead of "no significant effect".

Q33 – Q35: Went well

Q33, complete answer.

Because the coefficient on $hsize^2$ is negative, the turning point is a maximum – predicted score decreases after this point. $\partial sat/\partial hsize = 19.30 - 2(2.19) \cdot hsize = 0 \Rightarrow hsize^* = 19.30 / 4.38 \approx 4.41$.

Since hsize is in hundreds, 4.41 means about 441 students is the optimal size (440 also accepted).

Q33, common mistakes.

- Dropping the factor 2 in the derivative (giving 8.81, or 881 students). Converting units wrongly (440,000 students).
- Giving the number with no justification that it is a maximum.

Q34, complete answer.

The difference is the *black* coefficient alone, −169.81 points. With $t = -169.81/12.71 \approx -13.4 > 2$, it is significant at 5%. A nonblack male is expected to have higher score by 169.81 points than a black male.

Q34, common mistakes.

- Claiming significance without computing the t-statistic. No conclusion.

Q35, complete answer.

For females, both the *black* coefficient and the interaction apply: −169.81 + 62.31 = −107.50. A black female is expected to have lower score by 107.5 points than a nonblack female.

Q34, common mistakes.

- Leaving out the *female* or *black* term when building the black-female prediction. One such setup gave +107, with the right size but the wrong sign. No conclusion.

Q36: The concept is well understood (86% average, the highest of all open questions).
Most of the points lost came from parts b–d answered as numbers without the expression.

A complete answer.

of a house sold in 1978 not near the incinerator cite	82517
of a house sold in 1978 near the incinerator cite	82517-18824 = 63693
of a house sold in 1981 not near the incinerator cite	82517 + 18790 = 101307
of a house sold in 1981 near the incinerator cite	82517+18790-18824-11864 = 70619

Common mistakes.

- Giving the final number with no expression. The expression shows which differences you took and in which order.

Q37: Most students got the decision right and then overstated what it means

A complete answer. H_0 is homoskedasticity. With $p \approx 0.102$, we fail to reject H_0 at 1%, 5% and, just barely, 10%. So there is no significant evidence of heteroskedasticity (non-constant variance of errors), and the usual standard errors are likely acceptable.

Common mistakes.

- "The model is homoskedastic", "the variance is constant" or "MLR.5 is not violated". The test cannot show this.
- Reading the result backwards and concluding the model is heteroskedastic.
- Invented thresholds such as "F should be above 3". The p-value already gives the answer.
- "The model remains unbiased and consistent." Unbiasedness and consistency do not depend on MLR.5. What MLR.5 affects is the validity of the usual standard errors and whether OLS is BLUE.

Q38: Specification from the figure.
It averaged 19%, 59% of students scored zero, and even the top quarter averaged only 29%

A complete answer. Propose log_tfp = $\beta_0 + \beta_1 \cdot rd + \beta_2 \cdot exporter + u$. Exporters have both higher TFP and higher R&D, so leaving the dummy out would bias β_1. The relationship looks roughly linear, so no squared term is needed.

More points: log_tfp = $\beta_0 + \beta_1 \cdot rd + \beta_2 \cdot exporter + \beta_3(exporter \cdot rd) + u$. An exporter·rd interaction is reasonable to test since the slopes visibly differ.

More points: log_tfp = $\beta_0 + \beta_1 \cdot rd + \beta_2 \cdot exporter + \beta 3(exporter \cdot (rd - mean(rd)) + u$. The centred interaction keeps β_2 interpretable as the exporter premium at typical R&D intensity rather than at $rd = 0$.

Common mistakes.

- No equation at all. Typical answers proposed procedures or included the instruments instead of writing a specification justified by the figure.
- Adding dist_univ or tax_credit. The figure doesn't show them. These are the instruments in Q40–Q41, and putting them in the productivity equation contradicts the exclusion restriction.
- Deleting the outlier or giving it a dummy without first checking whether it is influential.
- Generic procedures ("check the MLR assumptions, then run 2SLS") instead of a specification.

Q39: Omitted variable bias

A complete answer. Management quality is unobserved, so it sits in u. Because it is correlated with *rd*, Cov(rd, u) ≠ 0, MLR.4 fails, and OLS is biased and inconsistent. This is omitted variable bias:

$$E(\tilde{\beta}_{rd}) = \beta_{rd} + \beta_{mgmt} \cdot \delta_1, \quad \text{where } \delta_1 = \text{Cov}(rd, mgmt) / \text{Var}(rd)$$

$\beta_{mgmt} > 0$ because management raises productivity, and $\delta_1 > 0$ because better-managed firms invest more in R&D. The bias is positive, so OLS *overestimates* the return to R&D.

Common mistakes.

- The wrong pair of signs: multiplying the effect of *rd* on productivity by the effect of management. The correlation between *rd* and management quality is what creates the bias.
- Only one link given, or the direction asserted with no argument.
- The wrong direction ("underestimated"), or the wrong quantity ("productivity is overestimated", "the effect of management is overestimated").
- Confusing OVB with multicollinearity: "inflated standard errors", VIF tests. OVB is about bias, not precision.
- Wrong assumption or object: MLR.6 (normality), Cov(log_tfp, u), or calling management quality the endogenous variable. *rd* is the endogenous regressor.
- "Double counting". Nothing is counted twice; the effect of management is wrongly attributed to *rd*.

Q40: Instrument relevance

A complete answer. t(dist_univ) = −0.052/0.004 = −13 and t(tax_credit) = 0.149/0.016 ≈ 9.31. Both |t| values far exceed 2, so Cov(z, rd) ≠ 0 for each instrument. Instruments are relevant. *(Ideally, mention a joint F-test of both instruments, with F > 10 as the rule of thumb for strong instruments, or a Cragg-Donald/Stock-Yogo check.)* Signs are as expected: firms further from a technical university do less R&D (weaker access to research collaboration and skilled graduates); firms in regions with more generous tax credits do more.

Common mistakes.

- No t-statistics: relevance asserted or argued only from economics.
- No economics: only t-statistics.
- Treating the negative sign as "not relevant". Cov(z, x) ≠ 0 holds for negative correlation too.
- Comparing −13 with −1.96 without the absolute value and calling it insignificant.
- Using 10 as the critical value for a t-test. The F > 10 rule applies to the first-stage F-statistic. For one instrument $F = t^2$, so tax_credit has F ≈ 87.
- Judging relevance by coefficient size ("0.052 is small, so dist_univ is weak"). With |t| = 13, dist_univ is the *stronger* instrument, and coefficient size depends on units.
- Arguing that small standard errors count against relevance. They are the reason the instruments are significant.
- Confusing relevance with exclusion, or concluding from the first stage that the instruments are "good" or "valid".

Q41: R^2 vs. 2SLS

A complete answer. No. As shown in Q39, *rd* is endogenous, so OLS is biased upward. A higher R^2 here reflects more contamination from the omitted variable, not a more accurate causal estimate.
2SLS uses only the variation in *rd* driven by the instruments and is consistent if they are valid. Its R^2 is lower by construction. A policy report on the returns to R&D needs to know what would happen if firms actually increased R&D spending – a causal question that only an unbiased estimate of β_1 can answer. So, since the OLS is biased, 2SLS with valid instruments is a better choice for a policy report.

Common mistakes.

- Choosing Model 1 because of its higher R^2.
- Comparing AIC, BIC or adjusted R^2. These are fit criteria too, and both models have the same regressors.
- There is no mention of why the R^2 is smaller in 2SLS.
- Saying the instruments cause bias, or that a lower R^2 means weak instruments. Q40 shows they are strong.
- Saying 2SLS reduces the sample size. It reduces the variation used, not the number of observations.
- Answering "no" without explaining why OLS fails here: no mention of endogeneity, bias or MLR.4.

Mistakes across all Part B questions

- **Failing to reject H_0 does not prove H_0.** Write "fail to reject H_0" and "no significant evidence of X", never "accept H_0", "the model is homoskedastic" or "the variable has no effect".
- **Hypotheses are about population parameters, not variables.** Write H_0: $\beta_{LINCOME} = \beta_{PRPPOV} = 0$, not "both variables are 0".
- **A p-value is not the probability that H_0 is true.** It is the probability of a test statistic at least this extreme if H_0 were true.
- **Significance level, not confidence level.** A test is run at the 5% significance level; a confidence interval is a different tool.
- **Show the evidence.** "Significant" needs a t-statistic or p-value next to it. If the question asks you to test, the calculation is part of the answer.
- **Use the right critical value.** A two-sided test at 5% uses 1.96 (about 2). The value 1.645 is one-sided, and 10 is the rule of thumb for a first-stage F-statistic, not for a t-statistic.

Recommendations for the rest of the course (1)

1. **Practise open questions, not just multiple choice.** The class averaged 81% on multiple choice but 56% on open questions, and the open questions decided most of the differences in final scores. Work through written problems where you have to explain, not just recognise the answer.
2. **Finish the argument.** Most points on Q30, Q37, Q39, Q40 and Q41 were lost by stopping after the first step. A complete answer has three parts: the decision (reject or fail to reject), what it means in this context, and why it happens. For example, "multicollinearity" alone is not an answer; explain that correlated regressors inflate the standard errors.
3. **Show your calculations and expressions.** Write the t-statistic, the derivative or the DiD expression before the number. On Q36, most students who lost points gave correct numbers without the expression.
4. **Treat endogeneity and IV as one connected story.** Q39–Q41 form a chain: why OLS is biased (omitted variable bias and the sign of the bias), whether the instruments are relevant (first-stage t- or F-statistics), and why 2SLS is preferred despite a lower R^2. This block cost about half of all lost points on the open questions, so it's the area to focus on.
5. **Base specifications on the evidence in front of you.** When asked to propose a model from a figure, write the equation and justify each term from what the figure shows. Don't add variables because they appear elsewhere in the question. Instruments, for example, do not belong in the main equation.

Recommendations for the rest of the course (2)

6. **Use precise testing language.**
   - Write "fail to reject H_0", never "accept H_0". Failing to reject does not prove that the errors are homoskedastic or that a variable has no effect.
   - State hypotheses about parameters ($\beta = 0$), not variables.
   - A p-value is not the probability that H_0 is true.
   - For a two-sided test at 5%, the critical value is 1.96 (about 2), not 1.645.
7. **Know which concept is which.**
   - Multicollinearity inflates standard errors; omitted variable bias biases the estimators.
   - Instrument relevance can be tested from the first stage; exogeneity cannot.
   - Statistical or economic significance is not the same as a large coefficient.
   - The error term, the residual and the standard error are three different things.
8. **Check units and signs before you finish.** Is the variable in hundreds, kilometres or percentage points? Does your answer have the right sign? Several answers had the right method but a final number that was 100 or 1,000 times too large, or had the wrong sign because a term was left out.
9. **Revise log interpretations until they are automatic.** Results on Q31 were all-or-nothing: students either knew the log-log elasticity or lost all the points. Practise one example each of log-log, log-level and level-log models.

Appendix

Midterm questions

Part A

(choose the correct answer)

1. Econometrics ...

a. Develops and uses statistical methods for estimating economic relationships
b. Studies the behavior of individual economic agents in making economic decisions
c. Deals with the structure, behavior, and decision-making of an economy as a whole
d. Applies mathematical methods to solve economic problems

2. The term '*u*' in an econometric model is usually referred to as the

a. Parameter
b. Hypothesis
c. Error term
d. Dependent variable

3. A data set that consists of a sample of individuals, households, firms, cities, countries, or a variety of other units, taken at a given point in time, is called a(n)

a. Panel data set
b. Time series data set
c. Pooled data set
d. Cross-sectional data set

4. In the equation $y = \beta_0 + \beta_1 x + u$, β_0 is the

a. Dependent variable
b. Intercept parameter
c. Independent variable
d. Slope parameter

5. If the total sum of squares (*SST*) in a regression equation is 90, and the residual sum of squares (*SSR*) is 66, what is the value of the R-squared?

a. 0.27
b. 0.73
c. 0.55
d. 1.56

6. The error term in a regression equation is said to exhibit homoskedasticity if _____

a. It has zero conditional mean
b. It has the same value for all values of the explanatory variable
c. It has a value of one given any value of the explanatory variable
d. It has the same variance for all values of the explanatory variable

7. Which of the following is a nonlinear regression model?

a. $y = \beta_0 + \beta_1 x^{1/2} + u$
b. $\log y = \beta_0 + \beta_1 \log x + u$
c. $y = 1 / (\beta_0 + \beta_1 x) + u$
d. $y = \beta_0 + \beta_1 x + u$

8. If an independent variable in a MLRM is an exact linear combination of other independent variables, the model suffers from the problem of _____

a. Homoskedasticity
b. Perfect collinearity
c. Heteroskedasticity
d. Omitted variable bias

9. The key assumption for the general multiple regression model is that all factors in the unobserved error term are correlated with the explanatory variables.

a. True
b. False
c. Not enough information to answer

10. High (but not perfect) correlation between two or more independent variables is called _____

a. Multicollinearity
b. Heteroskedasticity
c. Homoskedasticity
d. Micronumerosity

11. The normality assumption implies that:

a. The population error u is dependent on the explanatory variables and is normally distributed with mean equal to one and variance σ^2 .
b. The population error u is independent of the explanatory variables and is normally distributed with mean equal to zero and variance σ .
c. The population error u is independent of the explanatory variables and is normally distributed with mean equal to zero and variance σ^2 .
d. The population error u is dependent on the explanatory variables and is normally distributed with mean equal to zero and variance σ^2 .

12. Consider the equation: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$. $H_0: \beta_2 = 0$ states that:

a. x_2 has no effect on the expected value of β_2
b. β_2 has no effect on the expected value of y
c. y has no effect on the expected value of x_2
d. x_2 has no effect on the expected value of y

13. The significance level of a test is:

a. The probability of rejecting the null hypothesis when it is true
b. The probability of rejecting the null hypothesis when it is false
c. One minus the probability of rejecting the null hypothesis when it is false
d. One minus the probability of rejecting the null hypothesis when it is true

14. Which of the following tools is used to test multiple linear restrictions?

a. *t* test
b. *F* test
c. z test
d. Unit root test

15. If $\hat{\beta}_j$ is an unbiased and consistent estimator of β_j, then the:

a. Distribution of $\hat{\beta}_j$ becomes more and more loosely distributed around β_j as the sample size grows
b. Distribution of $\hat{\beta}_j$ becomes increasingly skewed as the sample size grows
c. Distribution of $\hat{\beta}_j$ becomes more and more tightly distributed around β_j as the sample size grows
d. Distribution of $\hat{\beta}_j$ remains unaffected as the sample size grows

16. In a regression model, if variance of the dependent variable, conditional on the explanatory variables, is not constant, then the:

a. t-statistics and confidence intervals are both invalid no matter how large the sample size is
b. t-statistics are invalid and confidence intervals are valid for small sample sizes
c. t-statistics are valid and confidence intervals are invalid for small sample sizes
d. t-statistics and confidence intervals are valid no matter how large the sample size is

17. In a regression model, which of the following will be described using a binary variable?

a. The volume of rainfall during a year
b. The percentage of humidity in air on a particular day
c. The concentration of dust particles in air
d. Whether it rained on a particular day or it did not

18. The following simple model is used to determine the annual savings of an individual on the basis of his annual income and education: $Savings = \beta_0 + \beta_1 Education + \beta_2 Income + u$. The variable $Education$ takes a value of 1 if the person is educated and the variable $Income$ measures the income of the individual. The benchmark group in this model is _____

a. The group of educated people
b. The group of educated people with high income
c. The group of uneducated people
d. The group of uneducated people with high income

19. The income of an individual in Studentia depends on his ethnicity and several other factors which can be measured quantitatively. If there are 5 ethnic groups in Studentia, how many dummy variables should be included in the regression equation (with an intercept) for income determination in Studentia?

a. 1
b. 4
c. 5
d. 2

20. Which of the following types of sampling always causes bias or inconsistency in the OLS estimators?

a. Endogenous sampling
b. Random sampling
c. Exogenous sampling
d. Stratified sampling

21. In the following equation, *gdp* refers to gross domestic product (in USD), and *fdi* refers to foreign direct investment (in % of GDP): $\log(gdp) = 2.65 + 0.527\log(bankcredit) + 0.222fdi$. Which of the following statements is then true?

a. If *gdp* increases by 1%, bank credit increases by approximately 0.527%, the level of FDI remaining constant
b. If bank credit increases by 1%, *gdp* increases by approximately 0.527%, the level of FDI remaining constant
c. If bank credit increases by 1%, *gdp* increases by approximately 0.527 USD, the level of FDI remaining constant
d. If bank credit increases by 1%, *gdp* decreases by approximately 0.527%, the level of FDI remaining constant

22. In the following equation, *gdp* refers to gross domestic product (in USD), and *fdi* refers to foreign direct investment (in % of GDP): $\log(gdp) = 2.65 + 0.527\log(bankcredit) + 0.222fdi$. Which of the following statements is then true?

a. If *fdi* increases by 1%, *gdp* increases by 22.2%, the amount of bank credit remaining constant
b. If *fdi* increases by 1 p.p. of GDP, *gdp* increases by 24.9%, the amount of bank credit remaining constant
c. If *fdi* increases by 1 p.p. of GDP, *gdp* increases by 26.5%, the amount of bank credit remaining constant
d. If *fdi* increases by 1%, *gdp* increases by 52.7%, the amount of bank credit remaining constant

23. Which of the following correctly identifies a limitation of logarithmic transformation?

a. Taking log of variables makes OLS estimates more sensitive to extreme values in comparison to variables taken in level
b. Logarithmic transformations of variables are likely to lead to heteroskedasticity
c. Logarithmic transformations cannot be used if a variable takes on zero or negative values
d. Taking log of variables often expands its range which can cause inefficient estimates

24. Which of the following models is used quite often to capture decreasing or increasing marginal effects of a variable?

a. Models with logarithmic functions
b. Models with variables in level
c. Models with interaction terms
d. Models with quadratic functions

25. Which of the following correctly identifies an advantage of using adjusted R-squared over R-squared?

a. The penalty of adding new independent variables is better understood through adjusted R-squared
b. Adjusted R-squared corrects the bias in R-squared
c. Adjusted R-squared is easier to calculate than R-squared
d. The adjusted R-squared can be calculated for models having logarithmic functions while R-squared cannot be calculated for such models

26. Consider the following simple regression model: $y = \beta_0 + \beta_1 x + u$. Suppose z is an instrument for x. Which of the following conditions denotes instrument exogeneity?

a. $\text{Cov}(z, u) \neq 0$
b. $\text{Cov}(z, x) \neq 0$
c. $\text{Cov}(z, u) = 0$
d. $\text{Cov}(z, x) = 0$

27. Consider the following simple regression model: $y = \beta_0 + \beta_1 x + u$. Suppose z is an instrument for x. Which of the following conditions denotes instrument relevance?

a. $\text{Cov}(z, x) = 0$
b. $\text{Cov}(z, u) = 0$
c. $\text{Cov}(z, u) \neq 0$
d. $\text{Cov}(z, x) \neq 0$

28. Consider the following regression equation: $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u$, where y is a binary variable. In this case, the estimated slope coefficient, $\hat{\beta}_1$ measures _____

a. The predicted change in the value of y when x_1 increases by one unit, everything else remaining constant
b. The predicted change in the probability of success when x_1 increases by one unit, everything else remaining constant
c. The predicted change in the value of y when x_1 increases by one percent, everything else remaining constant
d. The predicted change in the probability of success when x_1 increases by one percent, everything else remaining constant

29. The order condition for identification of an equation (2SLS) requires that there should be _____

a. At least one exogenous explanatory variable in a structural equation
b. At least as many dummy variables in an equation as there are exogenous explanatory variables
c. As many lagged independent variables in an equation as there are exogenous explanatory variables
d. At least as many excluded exogenous explanatory variables as there are included endogenous explanatory variables

Question	Correct answer
1	a
2	c
3	d
4	b
5	a
6	d
7	c
8	b
9	b
10	a
11	c
12	d
13	a
14	b
15	c
16	a
17	d
18	c
19	b
20	a
21	b
22	b
23	c
24	d
25	a
26	c
27	d
28	b
29	d

Part B

(write the answer)

30. Look at the EViews output below. Explain the Wald test results. What problem might this regression have and why?

Dependent Variable: LPSODA
Method: Least Squares
Date: 08/13/26 Time: 20:02
Sample: 1 410
Included observations: 401

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRPBLCK	0.097550	0.029261	3.333825	0.0009
LINCOME	-0.052990	0.037526	-1.412098	0.1587
PRPPOV	0.052123	0.134499	0.387532	0.6986
LHSEVAL	0.121306	0.017684	6.859600	0.0000
C	-0.841515	0.292432	-2.877644	0.0042
R-squared	0.183930	Mean dependent var		0.040299

Wald Test:
Equation: Untitled

Test Statistic	Value	df	Probability
F-statistic	3.522688	(2, 396)	0.0304
Chi-square	7.045376	2	0.0295

Null Hypothesis: C(2)=0, C(3)=0

The description and model below are applicable for tasks 31 and 32

Are rent rates influenced by the student population in a college town? Let *rent* be the average monthly rent paid on rental units in a college town in the United States. Let *pop* denote the total city population, *avginc* the average city income, and *pctstu* the student population as a percentage of the total population. One model to test for a relationship is (standard errors in parentheses):

$$\widehat{\log(rent)} = \underset{(.844)}{.043} + \underset{(.039)}{.066}\ \log(pop) + \underset{(.081)}{.507}\ \log(avginc) + \underset{(.0017)}{.0056}\ pctstu$$
$$n = 64,\ R^2 = .458.$$

31. What is wrong with the statement: "A 10% increase in population is associated with about a 6.6% increase in rent"? Correct the statement.

32. Use the rule of thumb to test at the 5% level: 1) the null hypothesis that the size of the student body relative to the population has no ceteris paribus effect on monthly rents against the two-sided alternative; 2) the null hypothesis that the total city population has no ceteris paribus effect on monthly rents against the two-sided alternative. What are your decisions on each test?

The description and model below are applicable for tasks 33–35

The variable *sat* is the combined SAT score, *hsize* is size of the student's high school graduating class (in hundreds), *female* is a gender dummy variable equal to one for women and zero otherwise, and *black* is a race dummy variable equal to one for blacks and zero otherwise. One model to test for a relationship between class size and academic performance is (standard errors in parentheses):

$$\widehat{sat} = \underset{}{1{,}028.10} + \underset{(6.29)}{19.30}\ hsize - \underset{(.53)}{2.19}\ hsize^2 - \underset{(4.29)}{45.09}\ female$$
$$- \underset{(12.71)}{169.81}\ black + \underset{(18.15)}{62.31}\ female{\cdot}black$$
$$n = 4{,}137,\ R^2 = .0858.$$

33. Using the estimated equation, what is the "optimal" high school size? Justify your answer.

34. Holding *hsize* fixed, what is the estimated difference in SAT score between nonblack males and black males? Is the difference significant at 5% level?

35. Holding *hsize* fixed, what is the estimated difference in SAT score between black females and nonblack females?

The description and model below are applicable for tasks 36–37

The following model uses the data on prices of houses pooled over two years – sold in 1978 and sold in 1981.

$$rprice = 82517 + 18790 * y81 - 18824 * nearinc - 11864 * y81 * nearinc$$

The variable *rprice* is the house price in real terms (1978 dollars). The variable *nearinc* is a binary variable equal to one if the house is near the incinerator, and zero otherwise. The variable *y81* is a binary variable equal to one if the house was sold in 1981.

The rumor that a new incinerator would be built began after 1978, and construction began in 1981. The hypothesis is that the price of houses located near the incinerator would fall relative to the price of more distant houses.

36. Using the model results above estimate the average price:

a) of a house sold in 1978 not near the incinerator _____

b) of a house sold in 1978 near the incinerator _____

c) of a house sold in 1981 not near the incinerator _____

d) of a house sold in 1981 near the incinerator _____

37. Look at the EViews output below. What can you conclude based on the test results?

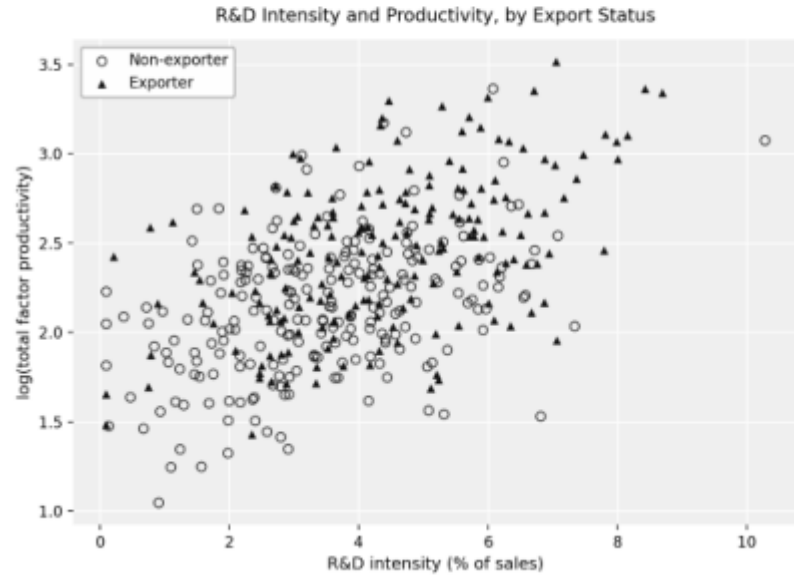
Heteroskedasticity Test: Breusch-Pagan-Godfrey
Null hypothesis: Homoskedasticity

F-statistic	2.081081	Prob. F(3,317)	0.1026
Obs*R-squared	6.199915	Prob. Chi-Square(3)	0.1023

The description and model below are applicable for tasks 38–41

A researcher studies 420 manufacturing firms to estimate the effect of R&D investment on productivity. She observes: *log_tfp* (natural log of total factor productivity), *rd* (R&D spending as % of sales), *exporter* (=1 if the firm exports), *dist_univ* (km from the firm to the nearest technical university), and *tax_credit* (the R&D tax credit rate, in %, offered by the firm's region).

38. Based on Figure below, what regression specification would you propose?



39. The researcher worries that *rd* is correlated with unobserved management quality, which independently raises productivity. Explain why this makes OLS estimates of the return to R&D unreliable, and predict the direction of the bias.

40. The researcher estimates the following first stage of the 2SLS procedure (standard errors in parentheses):

$$\underset{}{rd} = \underset{(0.19)}{4.07} - \underset{(0.004)}{0.052}\, dist_univ + \underset{(0.016)}{0.149}\, tax_credit + \underset{(0.13)}{0.78}\, exporter$$

What can you say about the relevance of the instruments (*dist_univ*, *tax_credit*)?

41. The researcher estimated the productivity equation two ways: (1) OLS; (2) 2SLS with *dist_univ* and *tax_credit* as instruments for R&D. Model (1) has a higher R^2 (0.32) than Model (2) (0.24). Does this make Model (1) the better choice for a policy report on the causal return to R&D? Explain.